

Summary:

Software Engineer with 3 years of experience in Embedded software development, eager to learn and grow in a dynamic environment. Passionate about solving complex technical problems, optimizing system performance, and delivering high-quality solutions.

Skills

Programming Languages:

- C, C++, Embedded C, CSS, C#, Java
- Data structures, STL and OOP.

Development environment and tools:

- Dev environment: Visual Studio, QT Creator, STM32 IDE, MPLAB
- Boot to Qt(B2Qt), Yocto Project, GDB
- Azure Repository, Azure DevOps, Service Now, Jira,
- Azure Cloud
- Beyond Compare, MobaXterm, SmartTTY, ST-Link

Embedded controllers:

- Hands-on working with PIC, STM32 microcontrollers
- System on Module (SOM)
- Communication protocols: UART, SPI, I2C, CAN, OneWire

Linux Internals and Networking:

- System calls, IPC mechanisms
- Network Programming using TCP and UDP sockets

Experience

Venkam Technologies Pvt Ltd, Bengaluru

Embedded Software Engineer

Aug 2024 – Jan 2025

- Implemented user stories and designed display screens using Qt Creator for medical device, integrating log management to enhance application functionality.
- Worked with B2Qt and Linux for embedded system development, ensuring seamless integration of software and hardware.
- Implemented battery management solutions using the I2C protocol, ensuring reliable and efficient power usage.
- Fixed software bugs, documented processes, and created detailed test case documentation for future reference.
- Utilized Jira for tracking progress, managing tasks, and maintaining clear documentation for project transparency.

Wipro

Dec 2021 – July 2024

Software Developer

- Developed and unit tested new user stories offerings prior to release for bug identification.
- Worked closely on troubleshooting and debugging the application.
- Utilize Send Grid, App Dynamics, Decibel Insight, and Azure services to monitor and generate reports of applications to pinpoint pain areas.
- Managed the stability, and maintenance of applications in the production environment.
- Patch software and deployed fixes to eliminate customer complaints and security problems.

Education

PES Institute of Technology (Visvesvaraya Technological University) - Shivamogga

2017 - 2021

Bachelor of Engineering in Electronics and Communications CGPA – 7.5

Class - XII (PCMCs), Vikasa Pre-University, Shivamogga (YOP - 2017)

Class - X, National Public School CBSE, 8.8 (YOP - 2015)

Certifications Trainings & Internships

Emertxe Information Technologies, Bengaluru

Embedded Software Engineer – Trainee

- Completed hands-on training as part of the **Embedded Software Engineer (QP ELE/Q1501)** certification under **Skill India / NSDC**.
- Gained practical experience in embedded systems development, microcontroller programming (PIC), device drivers, data structures, and Linux internals.

Hindustan Aeronautics Limited, Bengaluru – MRO division

- Gained insights into HAL's Maintenance, Repair, and Logistics (MRO) division, Helicopter operations, and Line Replacement Units (LRUs) used in the cockpit.
 - Microsoft Azure Fundamentals (AZ-900)
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Projects

STM32F407VG Bare-Metal Device Driver Development

- Designed and implemented low-level device drivers for USART, SPI, I2C, and RCC modules without using any third-party HAL libraries.
- Peripheral drivers (USART, SPI, I2C, RCC) are fundamental and widely used across embedded systems, demonstrating practical understanding of essential communication protocols
- Utilized direct register access to configure and control peripherals on the STM32F407VG microcontroller.
- Strengthened skills in embedded C programming, microcontroller architecture, memory mapped I/O and debugging at the register level.

LSB Image Steganography using C

- Developed a technique to encode a secret text file inside an image (BMP format) using the Least Significant Bit (LSB) encoding method.
- Implemented bitwise operations to hide the text within the image data, ensuring minimal changes to the image quality.
- Created an efficient decoding mechanism to extract the hidden text from the modified image using reverse bitwise operations. Strengthened expertise in working with file operations, pointers, and advanced C concepts such as bitwise manipulation and memory management.